

Handling Interactions

Objectives

- ◆ Handle user-initiated events
- ◆ Make components “remember” information with state
- ◆ Update the UI in two phases
- ◆ Update multiple state
- ◆ Update an object in state
- ◆ Update an array in state

Adding event handlers

- ◆ Handling events is similar to the way you handle events on DOM elements:
 - ◆ `onClick` instead of `onclick`. (camelCase)
 - ◆ `onClick={save}` instead of `onclick="save()"`

```
16 export function Hello() {  
17   | const sayHi = () => {  
18   |   alert("Hello everybody!!!");  
19   | }  
20   | return (  
21   |   <>  
22   |   | <button onClick={sayHi}>Hello</button>  
23   |   </>  
24   | );
```

```
<Button onClick={() => alert('Hello Everybody!!!')}>Hello</Button>
```

Event Handler Example (1)

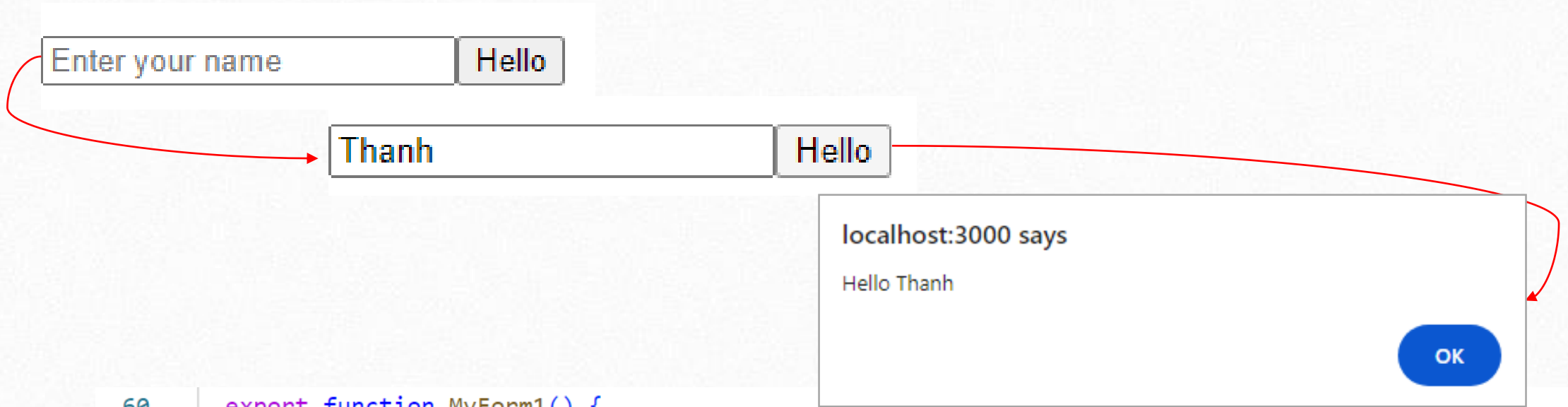
localhost:3000 says
Hello Everybody!!!

OK

```
58 export function MyForm1() {  
59   return (  
60     <div>  
61       <input type="text" placeholder="Enter your name"/>  
62       <button onClick={() => alert('Hello Everybody!!!')}>Hello</button>  
63     </div>  
64   );  
65 };
```

```
49 export function MyForm2({children}) {  
50   return (  
51     <div>  
52       <input type="text" placeholder="Enter your name"/>  
53       <button onClick={() => alert(children + ' Everybody!!!')}>{children}</button>  
54     </div>  
55   );  
56 };
```

Event Handler Example (2)



The diagram illustrates the state of a web application. It shows a form with a text input field and a button labeled "Hello". A red arrow points from the input field to a second state where the input field contains the text "Thanh" and the button still says "Hello". Another red arrow points from the "Hello" button to an alert dialog box. The alert dialog box displays the text "localhost:3000 says" and "Hello Thanh", with an "OK" button at the bottom right.

```
60 export function MyForm1() {
61   const [name, setName] = useState('')
62
63   return (
64     <div>
65       <input type="text" onChange={(e)=>setName(e.target.value)} placeholder="Enter your name"/>
66       <button onClick={()=>alert('Hello ' + name + '!!!')}>Hello</button>
67     </div>
68   );
69 }
```


State

- ◆ State is a component's memory
- ◆ Using a *useState* Hook to add state to a component
- ◆ *Hooks* are special functions that let your components use React features (state is one of those features)

useState Hook

- ◆ takes the initial state variable
- ◆ returns a pair of values: the current state, and a state *setter* function that lets you update it.

```
const [index, setIndex] = useState(0);  
const [showMore, setShowMore] = useState(false);
```

Exercise 01

1. Login Form
 - username = pass: alert thành công
2. Colors
3. Counter
4. Student Info
5. Clock

11:29:11 PM

My favorite color is red!

Blue Red Pink Green

My favorite color is blue!

Blue Red Pink Green

Counter App

Count: 15

Increment

Enter your name Enter your age ☐ Are you a student?

Name:

Age: 0

Student: No

Cát Hạnh 20 ☒ Are you a student?

Name: Cát Hạnh

Age: 20

Student: Yes

Exercise 02 - Menu

Dishes

- Spaghetti Carbonara
- Chicken Parmesan
- Tiramisu

Selected Dish

Select a dish to see details.

Dishes

- Spaghetti Carbonara
- Chicken Parmesan
- Tiramisu

Selected Dish

Spaghetti Carbonara



Pasta with bacon, eggs, and cheese
Price: \$12.99

Chicken Parmesan



Breaded chicken with marinara sauce and cheese
Price: \$14.99

Tiramisu



Italian coffee-flavored dessert
Price: \$6.99

Exercise 03

Next

***Floralis Genérica* by Eduardo Catalano**

(2 of 12)

Show details



```
name: 'Floralis Genérica',  
artist: 'Eduardo Catalano',  
description: 'This enormous (75 ft. or 23m) silver flower is  
url: 'https://i.imgur.com/ZF6s192m.jpg',  
alt: 'A gigantic metallic flower sculpture with reflective m
```

Next

***Floralis Genérica* by Eduardo Catalano**

(2 of 12)

Hide details

This enormous (75 ft. or 23m) silver flower is located in Buenos Aires. It is designed to move, closing its petals in the evening or when strong winds blow and opening them in the morning.



React batches state updates

- ◆ React waits until **all** code in the event handlers has run before processing your state updates.

```
export default function Counter() {  
  const [number, setNumber] = useState(0);  
  
  return (  
    <>  
      <h1>{number}</h1>  
      <button onClick={() => {  
        setNumber(number + 1);  
        setNumber(number + 1);  
        setNumber(number + 1);  
      }}>+3</button>  
    </>  
  )  
}
```



```
setNumber(0 + 1);  
setNumber(0 + 1);  
setNumber(0 + 1);
```


Updating Objects in State

```
116 export function StudentForm() {  
117   const [student, setStudent] = useState({  
118     id: '207PM47620',  
119     firstName: 'Thanh',  
120     lastName: 'Nguyen',  
121     email: 'thanh.207PM47620@vanlanguni.vn'  
122   });
```

ID:

First name:

Last name:

Email:

207PM47620 **Thanh Nguyen** *thanh.207PM47620@vanlanguni.vn*

```
124 function onFirstNameChange(e) {  
125   student.firstName = e.target.value;  
126 }
```

not recommended in React

```
128 function onLastNameChange(e) {  
129   setStudent({  
130     id: student.id,  
131     firstName: student.firstName,  
132     lastName: e.target.value,  
133     email: student.email  
134   });  
135 }
```

```
137 function onEmailChange(e) {  
138   setStudent({  
139     ...student,  
140     email: e.target.value  
141   });  
142 }
```

Object spread syntax

- ◆ update *only one* field in a form,
- ◆ keep the previous values for all other fields.

```
setPerson({  
  ...person, // Copy the old fields  
  firstName: e.target.value // But override this one  
});
```

- ◆ you can prepend an item by placing it *before* the original

```
setArtists([  
  { id: nextId++, name: name },  
  ...artists // Put old items at the end  
]);
```


Exercise 03

- ◆ Updating a nested object

```
const [person, setPerson] = useState({  
  name: 'Niki de Saint Phalle',  
  artwork: {  
    title: 'Blue Nana',  
    city: 'Hamburg',  
    image: 'https://i.imgur.com/Sd1AgU0m.jpg',  
  }  
});
```

Updating Arrays in State

- ◆ How to add, remove, or change items in an array in React state
- ◆ How to update an object inside of an array

Adding Item to an array

Item 4

- Item 1
- Item 2
- Item 3

```
setArtists( // Replace the state
  [ // with a new array
    ...artists, // that contains all the old items
    { id: nextId++, name: name } // and one new item at the end
  ]
);
```

```
setArtists(
  artists.filter(a => a.id !== artist.id)
);
```

Topics

1. React Router
2. React-Bootstrap
3. Tailwind CSS
4. JQuery
5. Redux
6. Angular